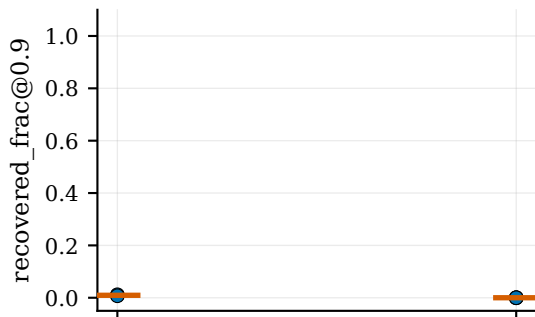


2×2 existence proof: orthogonality and stability are each necessary, jointly sufficient ($K = 32$). Note: i-strong is pool-restricted to $32 \leq d_{\text{state}}$.

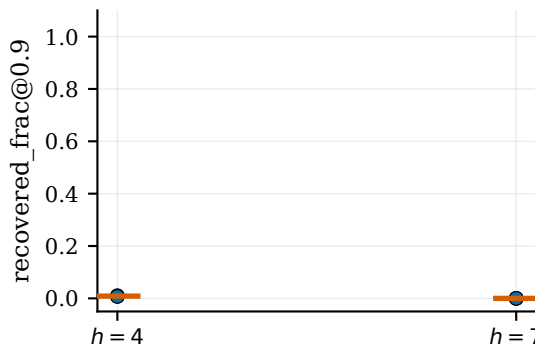
not-orthogonal, stable
(i (frozen), $n = 3$)



orthogonal, stable
(i-strong, $n = 3$)



not-orthogonal, not-stable
(iii-beta (baseline), $n = 3$)



orthogonal, not-stable
(bare geo3 (n12), $n = 3$)

